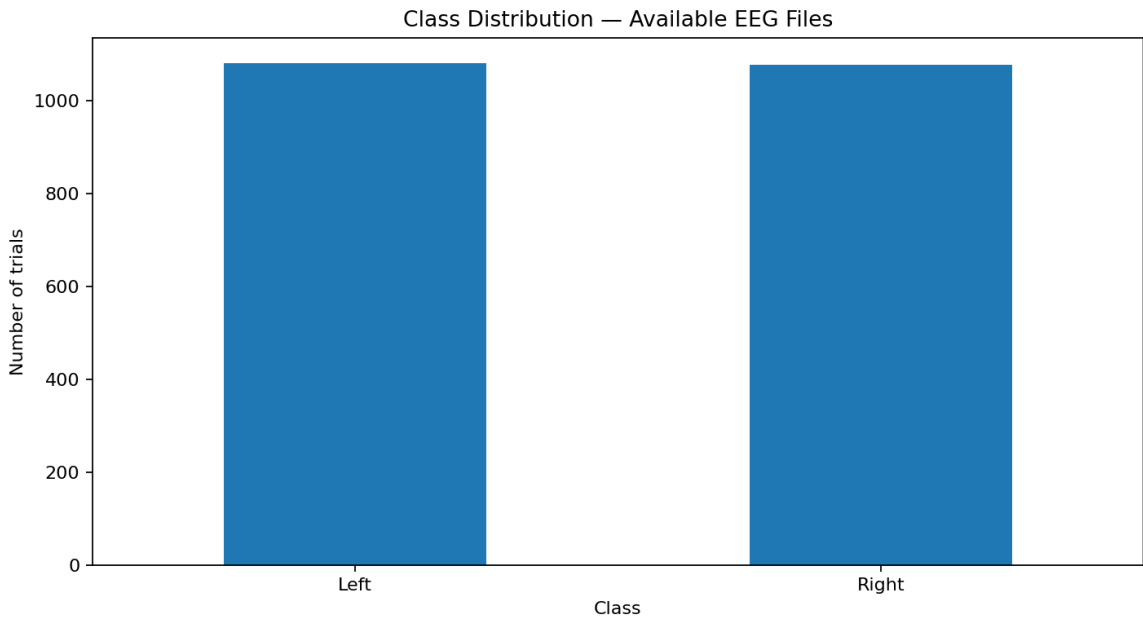


BCI Motor Imagery — Exploratory Data Analysis

Full analysis of the uploaded EEG CSV archive

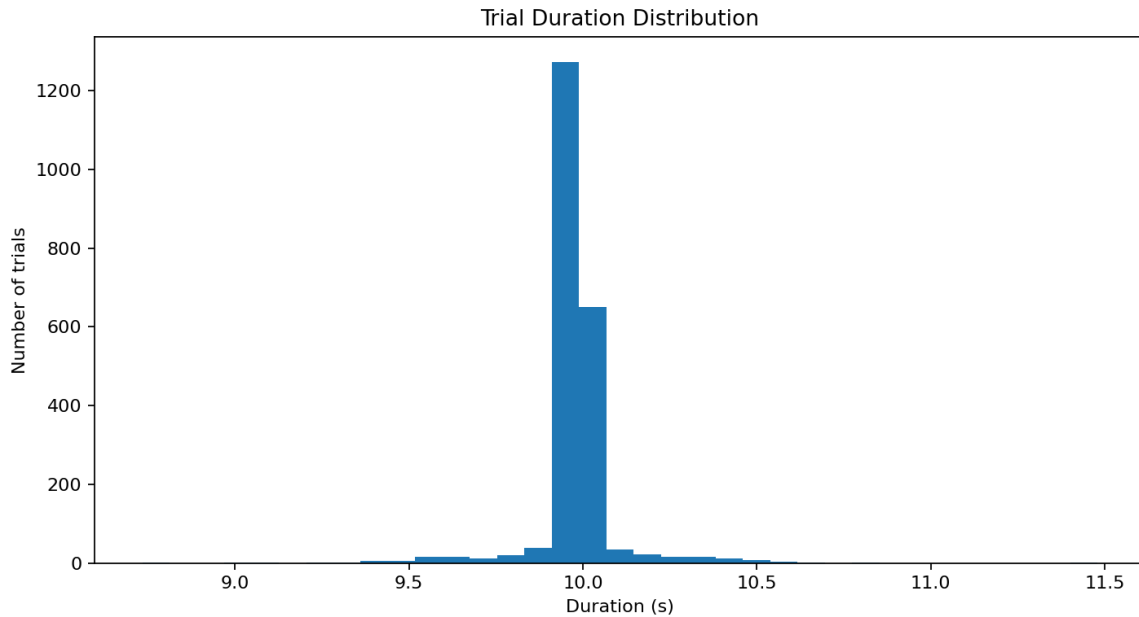
Important data note: Archive.zip contains 2,160 EEG CSV files (cellula_MI_data_1.csv through cellula_MI_data_2160.csv), while labels(2).csv contains 2,400 labels. Therefore this report analyzes the 2,160 available EEG files only, using the working assumption that file number *i* maps to label row *i*. The remaining 240 labels have no corresponding EEG file in the uploaded archive.

Item	Result
EEG files in archive	2,160
Labels supplied	2,400
Files analyzed	2,160
LEFT / Left	1,082 (50.09%)
RIGHT / Right	1,078 (49.91%)
Columns	Time, FZ, C3, CZ, C4
Samples per file	2,500 for all files
Median trial duration	9.985 s
Median inter-sample interval	2.497 ms
Median implied sampling rate	400.6 Hz
Files with NaN	0
Files with duplicate rows	0

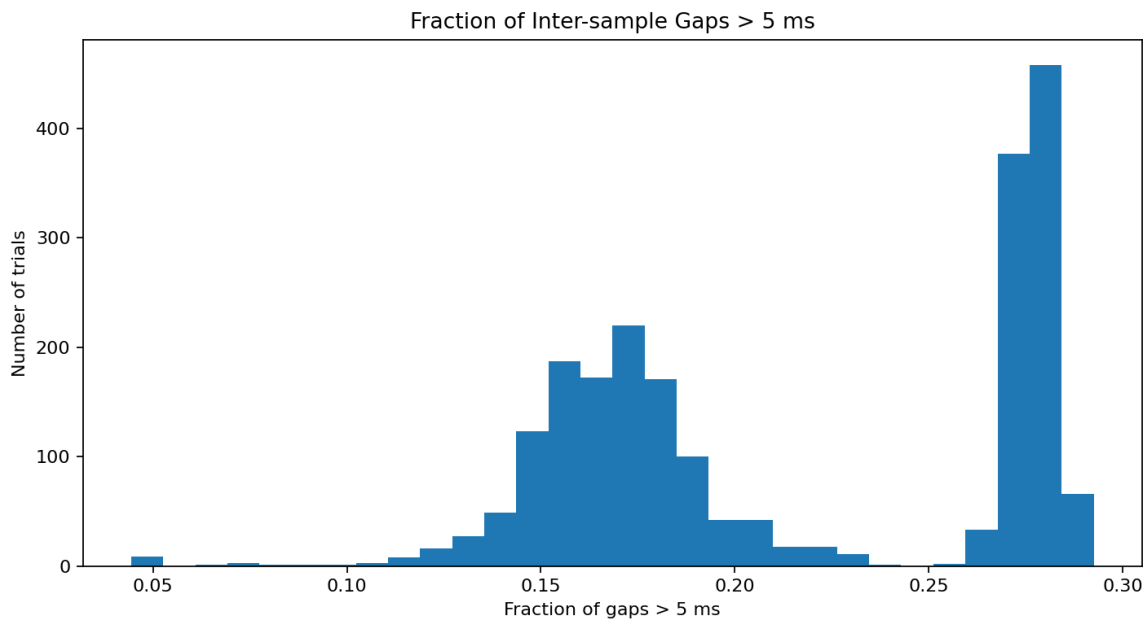


1. Trial Structure and Timing

All 2,160 available EEG files contain exactly 2,500 samples and the same five columns: Time, FZ, C3, CZ and C4. Trial duration has a median of 9.985 s, with a range of 8.734–11.480 s.

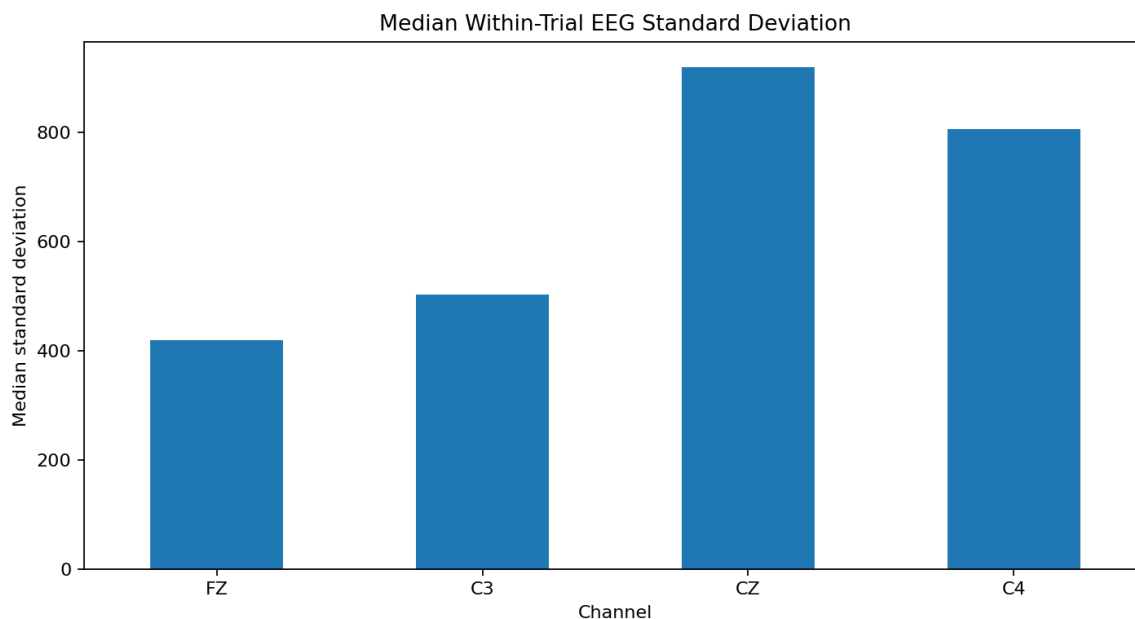


The timestamps are not uniformly spaced. The median inter-sample interval across trials is about 2.497 ms, corresponding to roughly 400.6 Hz, but many trials contain large timing gaps. On average, 21.5% of inter-sample intervals are >5 ms and 8.5% are >10 ms. Therefore frequency-domain analysis should not blindly treat the raw Time column as a perfectly regular sampling clock; a documented resampling step is recommended before Welch PSD/CSP processing.



2. Data Quality

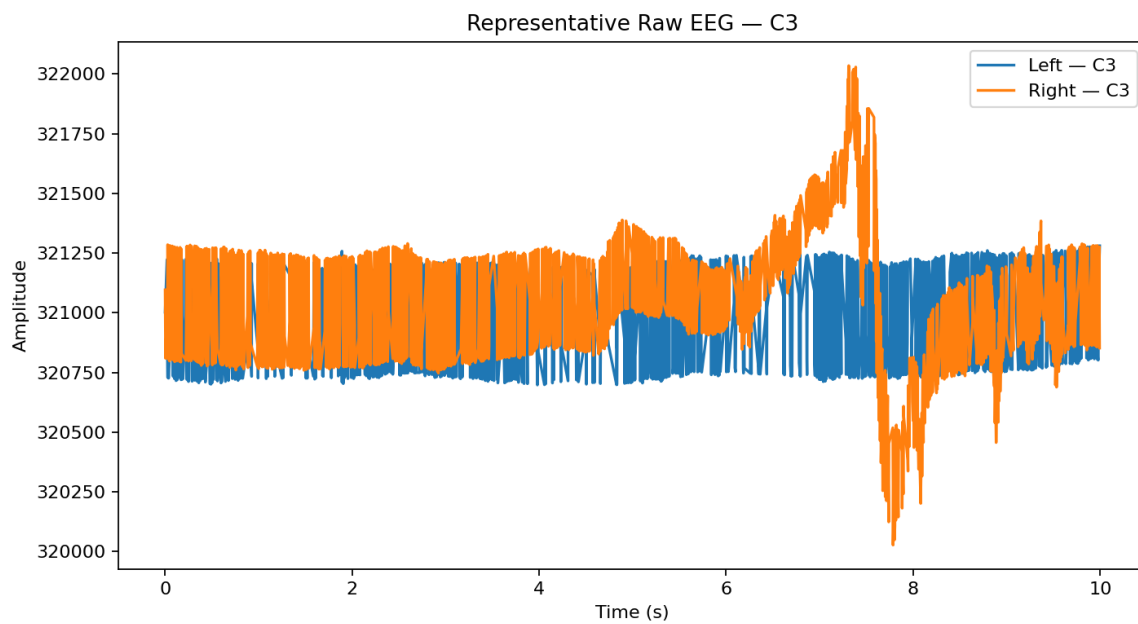
The available archive has no missing values and no duplicate rows in any of the 2,160 files. All files also have the expected five-column structure. However, amplitude variability is strongly heterogeneous across trials, so artifact screening is still required.



Median within-trial standard deviation is approximately FZ: 419.1, C3: 503.4, CZ: 920.0, C4: 805.9. Several trials have much larger variability than the median, which is consistent with the need for artifact/outlier screening before model training.

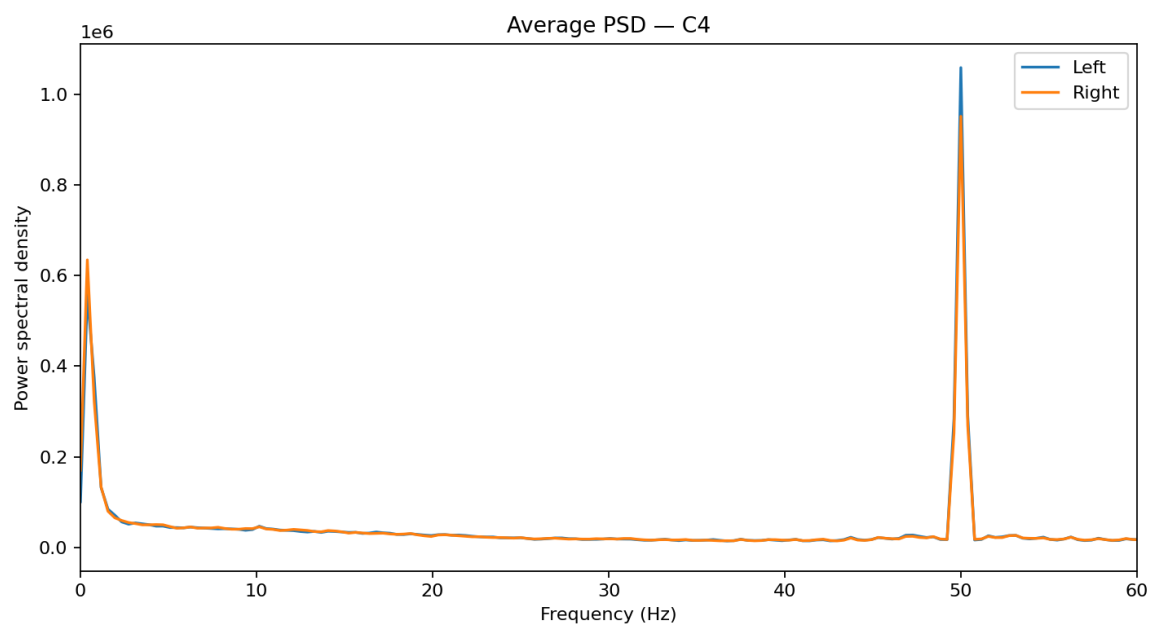
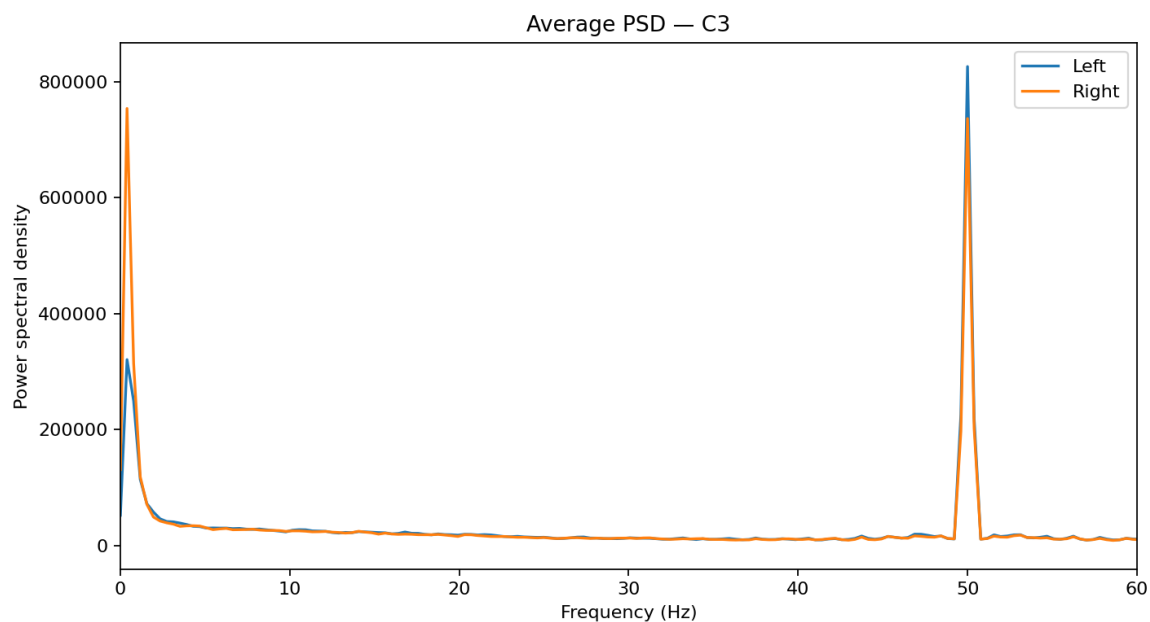
3. Raw EEG — Left vs Right

A representative raw C3 comparison is shown below. Raw amplitudes contain large offsets and are not the appropriate representation for direct LEFT/RIGHT classification. Filtering, detrending/re-referencing and epoching should precede feature extraction.

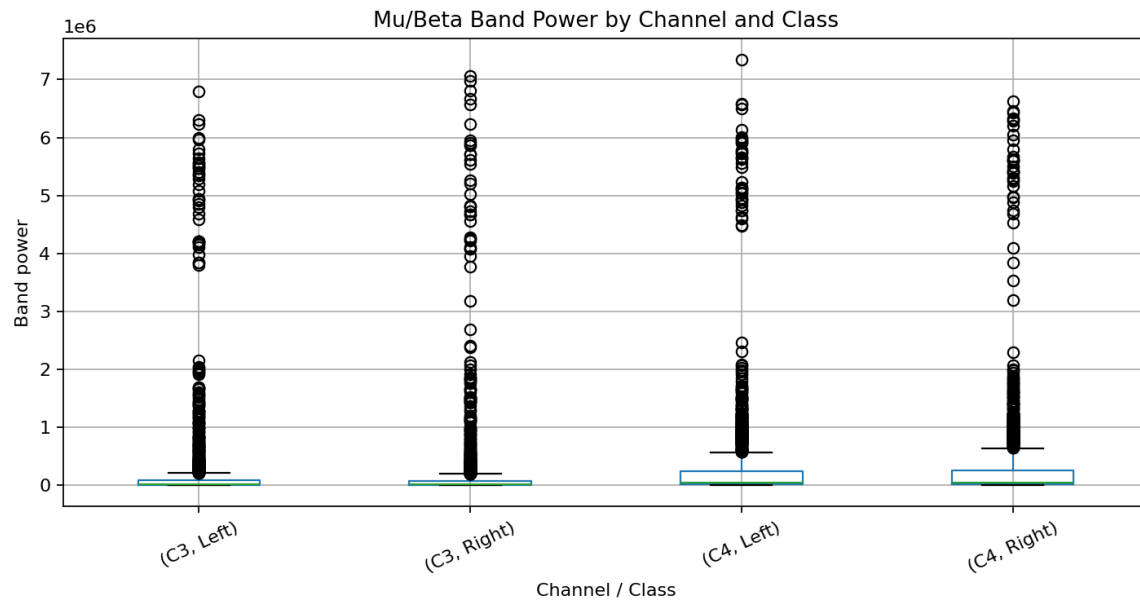


4. Frequency-Domain Analysis

Average PSD was computed after interpolating each irregularly sampled trial to a nominal 400 Hz grid and linearly detrending the signal. This 400 Hz choice is based on the robust median inter-sample interval (~2.5 ms), not on an explicit sampling-rate field in the CSVs. The project report identifies Mu (8–12 Hz) and Beta (13–30 Hz) as the key Motor Imagery ranges. ■filecite■turn0file0■



5. Mu and Beta Band Power



Class	Channel	Band	Median Power
Left	C3	Beta_13_30Hz	28614.06
Left	C3	Mu_8_12Hz	11506.70
Left	C4	Beta_13_30Hz	66804.74
Left	C4	Mu_8_12Hz	25193.49
Right	C3	Beta_13_30Hz	28539.08
Right	C3	Mu_8_12Hz	10461.16
Right	C4	Beta_13_30Hz	71247.24
Right	C4	Mu_8_12Hz	26655.34

6. Statistical LEFT/RIGHT Check

As an exploratory check, log10 band power was compared between Left and Right using a two-sided Mann–Whitney U test. These tests are descriptive only; they do not replace subject-aware validation and should not be interpreted as proof of a subject-independent biomarker.

Channel	Band	p-value
C3	Mu_8_12Hz	0.2290
C3	Beta_13_30Hz	0.3488
C4	Mu_8_12Hz	0.9744
C4	Beta_13_30Hz	0.7542

Interpretation

In the available 2,160 trials, the marginal C3/C4 Mu and Beta band-power distributions do not show a strong LEFT/RIGHT separation by this simple trial-level test. That does **not** mean the dataset cannot support Motor Imagery classification: CSP uses spatial covariance structure, and deep models can exploit temporal/spectral patterns beyond a single band-power summary. It does mean that the EDA does not provide evidence for a simple one-feature classifier based only on Mu/Beta power.

7. Recommended Next Steps

1) Confirm the label-to-file mapping and obtain the missing 240 EEG files. 2) Confirm the true acquisition sampling rate from the dataset documentation/device metadata. 3) Resample to a documented uniform rate, remove baseline/DC components, apply the project's band-pass and a 50/60 Hz notch as appropriate, then perform artifact screening. 4) Define a consistent task epoch and baseline. 5) Re-run PSD/ERD specifically around C3/C4. 6) Train the recommended CSP + SVM baseline first, then EEGNet. 7) Use subject-aware or trial-aware validation as emphasized in the project report. The report explicitly recommends starting with a clean CSP+SVM baseline before more complex deep models. ■filecite■turn0file0■